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RESEARCH ARTICLE

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A realistic MAP/PH/c queuing model for electric vehicle charging stations: theoretical development and real-world case study

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ABSTRACT

The growing demand for electric vehicles has been accompanied by increasing challenges related to queuing congestion and waiting times, especially during peak travel periods. Thus, understanding the variability and dynamics of queuing behavior during charging is critical for congestion management and charging supply planning. This paper details a realistic multi-server queuing model for electric vehicle (EV) charging stations in which multiple identical connectors operate independently. The arrival of EVs is modelled as a Markovian arrival process (MAP), with charging durations following a phase-type (PH) distribution. The findings reveal a lower bound of charging probability beyond which the queuing system is stable. The model is further validated via numerical examples and a case study using real-world data from over 300,000 charging sessions. This advanced stochastic framework provides theoretical and practical insights that can enable more efficient EV charging station design, management, and user satisfaction. Furthermore, the developed stochastic queuing process provides demand references that can inform the design of EV infrastructure capable of meeting growing EV adoption while ensuring a seamless user experience.

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Electric vehicle charging station; EV infrastructure queuing models; markovian arrival process (MAP); phase-type (PH) distribution; MAP/PH/c queuing model

1. Introduction

The increasing adoption of electric vehicles (EVs) has created new challenges for charging station operations. Congestion management and minimising waiting times are prominent among these challenges, as long queues and delays frustrate users and reduce overall satisfaction with charging experiences. Peak travel periods such as holidays and long weekends have proven especially problematic, as EV drivers have reported significant difficulties in finding available chargers and excessively long recharge times. For instance, during the Christmas season in the UK, Tesla owners reported waiting up to six hours to charge their vehicles, with queues of nearly 24 Teslas observed at a single station (LADbible-News 2022). Similarly, during China's National Day holiday, EV owners encountered extended waits at charging stations due to EV adoption outpacing infrastructure development (CNEV-Post 2021). This infrastructure lag has also been documented in Australia, with EV drivers reporting long queues and non-functional chargers during their summer road trips (ABC-News 2025). Such challenges serve as barriers to EV adoption and highlight the importance of modelling EV charging demand in networks more effectively to meet the needs of EV users, particularly during peak travel times.

To address these issues, it is essential to use realistic queuing models that accurately capture the inherent variability in EV arrival and charging times, thereby enabling the planning and optimisation of EV charging stations. While traditional models, such as M/M/c, have been used extensively to study queuing systems in recent years, such models rely on basic assumptions, such as arrivals following a Poisson process and service times being exponential. These assumptions do not reflect actual behaviour at EV charging stations, where arrivals are often correlated to common scheduled commuting patterns or other factors, and service times vary based on vehicle type and battery condition. Simplified or unrealistic models can underestimate congestion and

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wait times, resulting in poor infrastructure planning. Thus, more advanced models capable of capturing the underlying dynamics of EV charging systems are required to address these challenges.

This study employs the Markovian arrival process (MAP) and phase-type distribution (PH) to model EV queuing behaviours during charging. Unlike Poisson-based models, which assume that independent arrivals and service times follow simple Poisson and exponential distributions, MAPs and PH distributions capture the dynamics and autocorrelation of queuing processes in more realistic scenarios. For example, the MAP has been effectively applied to capture overdispersed and positively correlated arrival counts in call centre data modelling (Bernal, Lillo, and Ramirez-Cobo 2024), to model non-renewal arrival processes in telecommunication (Buchholz 2003), and to approximate the long-range dependence of traffic traces (Casale, Zhang, and Smirni 2010). MAPs can also represent scenarios wherein demand increases during specific periods, such as morning or evening commutes (Bernal, Lillo, and Ramirez-Cobo 2024), a feature which has been documented repeatedly in the literature focusing on EV arrival patterns, see Lahariya, Benoit, and Develder (2020) and Jonas, Daniels, and Macht (2023) among others. PH distributions, on the other hand, capture the stochastic behaviour of service times. This flexible approach has been employed successfully to model service times in different applications, from call centres (Kim et al. 2013) to inventory problems (Anbazhagan et al. 2022). Together, these models enable more accurate representations of system behaviour, particularly in stochastic systems, and they are adaptable to different types of variability in solving real-world charging and queuing problems for electric vehicles.

The rest of the paper is organised as follows: Section 2 provides a review of the literature relating to EV charging queuing and stochastic modelling; Section 3 details the proposed stochastic EV queuing model based on MAP and PH distributions; Section 4 examines a selection of case studies and uses real-world datasets to validate the proposed model; Section 5 discusses the applications and implications of the proposed model in EV charging management and infrastructure planning; and Section 6 provides some concluding remarks regarding the findings and avenues for future research. For consistency across sections, all mathematical symbols employed in the modelling and analysis are summarised in Table 1.

2. Literature review

2.1. Stochastic EV queuing and simplified assumptions

Prior works have explored EV charging and discharging coordination using system-level approaches focusing on optimisation, pricing, charging behaviours and energy management contexts (Ahmed et al.

Table 1. List of notations used in this paper.

Symbol	Definition
$I_a(t)$	Continuous-time Markov chain of the underlying MAP
m_a	Number of phases in the embedded MAP
D_0	MAP sub-generator matrix representing transitions without arrivals
D_1	MAP matrix representing transitions that include arrivals
$D = D_0 + D_1$	Infinitesimal generator of $I_a(t)$
α	Initial phase distribution of the MAP arrival process
θ	Stationary probability vector of the MAP
$\lambda = \theta D_1 e$	Effective arrival rate of the MAP (vehicles/hour)
$I_s(t)$	Continuous-time Markov chain of the underlying PH-distribution
m_s	Number of phases in the embedded PH-distribution
S	PH sub-generator matrix representing transition rates within transient service phases
$T = -S\mathbf{1}$	Exit rate vector of the PH distribution
β	Initial phase distribution of the PH service-time distribution
$\mu = -1/(\beta S^{-1} e)$	Mean service (charging) rate of the PH distribution (vehicles/hour)
$X(t)$	Continuous-time Markov chain of the queue
$N_q(t)$	Total number of EVs in the station at time t
Q	Infinitesimal generator of the process $X(t)$
c	Number of charging connectors (servers)
ρ	Utilisation factor, $\rho = \lambda/(c\mu)$ (dimensionless)
L_q	Expected queue length in steady state (vehicles)
W_q	Expected waiting time in steady state (hours)
π	Stationary probability vector of the MAP/PH/c CTMC
π_0	Steady-state probability that all connectors are idle
e	Column vector of ones of appropriate dimension

2016; Chekired et al. 2017; Alghamdi, Said, and Mouftah 2019; Javanmard et al. 2024; Neigum et al. 2025; Javanmard and Tang 2026). Although complementary to these studies, the present work addresses a gap in the literature by focusing on stochastic queuing analysis, rather than operational control or scheduling mechanisms.

Existing EV queuing behaviour studies largely utilise traditional frameworks (e.g. M/M/c) with finite or infinite queue capacities, with a focus on reducing waiting times, minimising costs, optimising station placement, or balancing energy distribution (Li and Zhang 2012; Said, Cherkaoui, and Khoukhi 2013; Said and Mouftah 2020; Zhang et al. 2021). Early efforts extended these models to the M/M/∞ case, incorporating random interruptions due to grid instability (Baek et al. 2011). Similarly, EV-battery swapping services have been modelled using M/D/c queues with deterministic swapping times (Tan, Sun, and Tsang 2014), and priority mechanisms for EV charging services that use cloud communication between EVs and stations have been integrated into M/M/c queues (Eldjalil and Lyes 2017).

Recently, several optimisation models have been developed, primarily using M/M/c and M/G/c queuing models, to determine the optimal locations and capacities of charging stations designed for intercity travel, electric taxis, and ride-hailing services (Jung et al. 2014; Yang, Dong, and Hu 2017; Wang et al. 2019; Chen, Lei, and Ukkusuri 2024; Wu et al. 2024). Other recent interactive studies have improved conventional queuing models by optimising the operating costs of EV charging stations via the integration of demand-responsive pricing and a combination of mobile and fixed charging units (Yang et al. 2021; Deng, Chen, and Lin 2025).

Existing approaches mainly rely on Poisson arrivals or simplified service-time distributions (e.g. exponential, deterministic, or general) for EV charging and queuing. However, these assumptions do not capture the bursty, correlated nature of real-world EV arrivals and the time-varying charging processes. Numerous researchers, including Zhang and Grijalva (2015), Farkas and Telek (2018), and An et al. (2025), have emphasised that Poisson-based models rely on unrealistic assumptions and lack proper validation, thus resulting in queuing models that under- or over-estimate station congestion.

The present study aims to address these gaps by leveraging the MAP and PH distributions to develop a realistic MAP/PH/c queuing model for EV charging stations. The proposed model accounts for the complexities of real-world conditions and offers an innovative approach to analysing and improving charging station performance. By addressing the limitations of traditional frameworks, this work provides a practical and efficient solution to reduce congestion and enhance the overall functionality of EV infrastructures.

2.2. Time-dependent stochastic queuing modelling in EV charging

In recent years, some progress has been made in developing advanced methods to tackle the time-varying charging process of EVs (Yousuf et al. 2024; Martífnez-Hernández, Tang, and Almehdawe 2024; Soufiani, Almehdawe, and Tang 2025). In one such work, Zhang and Grijalva (2015) employed nonhomogeneous Poisson arrivals and established a general charging time distribution to develop an M*t*/G/∞/N*max* model that accounts for time-varying arrival rates in small residential communities. Dong et al. (2016) introduced a DBMAP/PH/N/R queuing model in which charging durations followed a PH distribution and plug-in hybrid EV (PHEV) arrivals were represented by a discrete-time batch Markovian arrival process (DBMAP). The authors sought to develop a battery replacement strategy capable of enhancing charger efficiency and reducing waiting times. Their framework employed a discrete-time approach for the arrival process and a dimension reduction technique (also described in He (2014)), to minimise the size of transition blocks and overall state space. Later, Farkas and Telek (2018) constructed a MAP(2)/G/c model for charging station capacity planning by fitting a second-order MAP to the arrival process. However, their model focused on a specific scenario with a fixed number of arrival phases and a general distribution for service times, rather than on broader applications.

Some EV-infrastructure studies have sought to move beyond simplified assumptions about charging times by relaxing the exponential service time constraint to enable greater flexibility. Fan, Sainbayar, and Ren (2015) developed an M/G/c model fast charging DC stations for Li-ion batteries capable of varying charging times due to state-of-charge (SoC)-dependent power dynamics. Elsewhere, Gnann et al. (2018) built an M/G/1 queuing model to forecast future fast charger demand by estimating optimal vehicle-to-charger ratios and incorporating normally distributed service times. More recently, Pourvaziri et al. (2024)

successfully applied a GI/GI/c model alongside a deep learning (DL) model to achieve greater flexibility in forecasting demand and reducing waiting times.

While recent advances in DL and reinforcement learning (RL) have yielded powerful tools for forecasting charging demand and supporting real-time scheduling problems, these methods and the MAP/PH/c framework proposed in this work address fundamentally different problems. DL models can predict arrival rates or charging demand, and RL models can generate adaptive scheduling policies, but neither approach can produce a queueing model of the system. When imported into a queueing context, these models' forecasts and policies typically yield general or time-varying Markovian queueing systems (such as GI/G/c or Mt/G/c). Since such systems typically do not contain closed-form expressions for performance measures, their analysis is often dependent on simulations or numerical approximations. In contrast, the proposed MAP/PH/c formulation provides a structured, interpretable, and analytically tractable representation of arrival dependencies and service-time variability. Each phase and transition rate in the MAP and PH components has a clear operational meaning, which enables performance metrics to be calculated with a precision not possible with learning-based methods. Therefore, the MAP/PH/c approach complements data-driven forecasting and scheduling tools by providing an interpretable analytical framework and measurable system outputs that are crucial for long-term planning and congestion management.

2.3. Markovian arrival process (MAP) and phase-type (PH) distribution in service systems and EV charging

MAP and PH models and their underlying concepts have been explored across several service domains, including bursty IP traffic and superposed ATM cell streams (Kang et al. 2002; Chen et al. 2021). In addition, researchers have demonstrated these models' utility for capturing overdispersed and correlated arrivals in call centre applications (Bernal, Lillo, and Ramirez-Cobo 2024), and they have applied MAPs to characterise workloads in cloud computing, effectively modelling the static and dynamic correlations of IT traffic (Pacheco-Sanchez et al. 2011). MAP/PH models build on these foundations to enable the modelling of a variety of queuing contexts, thus allowing them to capture the complexities of uni-server and multi-server systems more accurately. These models have incorporated customer behaviours and system dynamics, such as impatience (with or without system failures), retrial behaviour, priority handling, and capacity constraints. For example, Chakravarthy (2009) applied an MAP/M/1 model to develop a disaster-recovery framework for telecommunication systems. Under normal operations, there was no impatience or capacity limitation; however, during failure repairs, impatient customers and new arrivals were redirected to a finite buffer of size K to ensure functionality while managing impatience. In a different work, Dudina, Kim, and Dudin (2013) examined a retrial and impatient system in a call centre context wherein customers unable to wait in the buffer would retry later. The authors used alternative techniques (described in Klimenok (2006)) to prove ergodicity and compute stationary distributions. Conversely, Chakravarthy (2013) analysed retrial behaviour using phase-type distributions while conducting simulations to evaluate performance metrics.

Priority handling and heterogeneous customer classes have also been explored within MAP/PH/c research. For instance, Kim et al. (2013) employed a MAP/PH/N/N+R framework to model call centre systems with priority and non-priority customers. The authors split MAP representations into distinct arrival transition rates and incorporated impatient time distributions, thus enabling stationary and waiting time distributions and performance evaluation to be derived under varying correlation levels and priority configurations. Furthermore, capacity and reliability considerations have also been central to MAP/PH/c models and studies. Asmussen and Møller (2001) analysis of steady-state waiting time distributions in MAP/PH/c systems provided a foundational framework for reliability applications in multi-server environments. In reliability engineering, MAP/PH/c models have been applied to capture correlated inter-failure times for electrical components, thus yielding valuable insights for maintenance optimisation (Rodríguez, Lillo, and Ramírez-Cobo 2015). Indeed, the integration of MAP/PH models within inventory systems has facilitated analyses of customer and inventory interactions and cost-function optimisations, and enabled practical design insights for systems operating in dynamic environments (Anbazhagan et al. 2022). The present study represents the joint probability distribution of customer counts and inventory levels in steady state and derives sojourn-time distributions using the Laplace-Stieltjes transform.

EV charging stations and infrastructure can also be regarded as service systems wherein users arrive in place, receive charging services, and depart afterwards. Previous studies applying MAP and PH models to

diverse service domains have demonstrated their robustness in addressing the stochastic complexities of modern systems, including EV charging and queuing. In this work, we leverage the strengths of MAP and PH distributions to establish a realistic MAP/PH/c queuing model for EV charging stations. In addition to accounting for the complexities of real-world conditions, our model provides an innovative approach to analysing and improving charging station performance. Thus, the proposed model addresses the limitations of traditional frameworks while providing a practical and efficient solution for reducing congestion and enhancing the overall functionality of EV infrastructures. A comparative summary of key queuing models relating to EV charging analysis is provided in Table 2.

2.4. Motivation and contributions

As demand grows, EV charging and queuing exhibit complex stochastic behaviours, time-varying dynamics, and correlations that are not well captured by traditional frameworks and assumptions of independent arrival and departure processes. The literature highlights advances in the evolution of MAP/PH models for analysing diverse system dynamics in service domains ranging from customer behaviour and reliability engineering to telecommunications and inventory management. This underscores the need for specialised queuing models that can accurately capture the complexities of modern EV charging behaviours and their infrastructures.

To the best of our knowledge, this study made one of the first attempts to develop and validate a comprehensive continuous-time MAP/PH/c queuing model specifically for electric vehicle (EV) charging systems using real-world data. Although continuous-time queuing models such as M/M/c and M/G/c have been previously applied for EV charging analysis, these frameworks do not account for the correlated and bursty nature of EV arrivals or the heterogeneous variability of charging durations observed in practice.

Furthermore, our model contributes detailed mathematical formulations of the MAP/PH/c structure, including the explicit representation of stochastic matrices, the derivation of steady-state probabilities, proof of stability conditions, and the computation of performance measures within the operational context of EV charging stations. Collectively, these elements establish a rigorous and data-validated analytical foundation for understanding and evaluating queuing behaviour in real EV charging environments.

Beyond theoretical contributions, this work provides practical insights based on the application of the model to real-world EV charging data. The case study indicates how the MAP/PH/c model effectively captures the complexities and inherent variability of real-world EV charging dynamics. Specifically, we fit the MAP and PH distributions to empirical arrival and charging time data to demonstrate the model's ability to address critical operational challenges such as congestion management and resource allocation.

This study highlights the proposed framework's robustness in meeting the necessary queuing requirements, thus reinforcing its potential as a reliable tool for planning and optimising EV charging infrastructure.

Table 2. Comparison of queuing models for EV charging analysis.

Category/Feature	M/M/c	M/G/c	BMAP/PH/N/R	MAP/PH/c
Arrival Process Representation				
Memoryless arrivals	Yes	Yes	No	No
Correlation/burstiness	No	No	Yes (batching)	**Yes (general correlation)**
Time-dependent intensity	No	No	Limited	**Yes (MAP structure)**
Service-Time Representation				
Distribution flexibility	Low	High	High	High
Captures EV charging variability	Low	Moderate	High	**High (SoC-phase interpretation)**
Empirical fitting suitability	Poor	Limited	Moderate	**High**
Interpretability and Behavioural Meaning				
Parameter interpretability	High (simple)	Moderate	Moderate	**High**
Links to observable charging behaviour	No	No	Limited	**Yes**
Operational relevance	Low	Moderate	Moderate	**High**
Analytical Tractability				
Closed-form performance metrics	Yes	Partial	Partial	**Yes**
Scalability to multi-server setting	Yes	Yes	Yes	Yes
Computational tractability	High	Moderate	Moderate-High	High (matrix-analytic)
Suitability for EV Charging Systems				
Captures real EV arrival patterns	Low	Low-Moderate	Moderate-High	**High**
Captures heterogeneous charging times	Low	Moderate	High	**High**
Overall realism for EV stations	Low	Moderate	Moderate-High	**High**

3. Problem statement of the EV charging process and queuing

To model the operation of an EV charging station, we consider a queuing system with c identical connectors, each working independently. The station operates on a first-come, first-served basis, with no special priority given to any vehicle and no impatience from customers. When an EV arrives at the station and finds an available connector, it starts charging immediately. If all the connectors are occupied at the time of arrival, the EV joins the queue and waits for its turn. Additionally, we assume that the station has unlimited capacity in its parking lot, allowing all arriving EVs to charge or queue without special restrictions. Although parking lots have limited capacity, we assume that the available capacity can accommodate all waiting vehicles. Once the charging process is completed, the EV leaves the station immediately. The following details the mathematical representation of the arrival and service processes that define this system.

3.1. Arrival and service time process

In the literature, it is well documented that the arrival of EVs at charging stations exhibits correlated and bursty patterns, often influenced by driver behaviour, traffic conditions, and the time of day (Dong et al. 2016; Farkas and Telek 2018). To model these complexities, we use a Markovian Arrival Process (MAP), a doubly stochastic process $\{N(t), I_a(t), t > 0\}$, where $I_a(t)$ represents an irreducible continuous-time Markov chain (CTMC) with finite state space $\{1, 2, \ldots, m_a\}$, and $N(t)$ is a counting process that records the total number of arrivals up to time t. The transition dynamics are governed by two rate matrices, D_0 and D_1, where D_0 describes transitions without arrivals and D_1 accounts for transitions that include arrivals. By definition, $D = D_0 + D_1$ serves as the infinitesimal generator matrix of the underlying Markov process $I_a(t)$. For further details on MAP, we refer the reader to He (2014).

Similar to EV arrivals, EV charging durations exhibit significant variability (Dong et al. 2016), driven by a combination of technical and environmental factors. A key factor is the state of charge (SoC) of the arriving EV, as vehicles with nearly empty batteries require significantly longer charging times than those with a higher SoC. Larger battery capacities also contribute to extended charging durations, as these batteries naturally require more time to replenish. Furthermore, delays can worsen when EVs designed for high-power charging are connected to lower-capacity chargers, resulting in slower energy transfer and longer queues. Environmental conditions, such as extreme heat or cold, further influence charging times, as battery protection mechanisms reduce charging rates to prevent overheating or freezing, ensuring safety and longevity. The useful characteristics of the PH-distribution can assist in realistically modelling the charging duration of EVs by incorporating the underlying Markov chain.

According to Neuts (1981), the PH distribution is the distribution of time until absorption (charging completion in our model), in a CTMC, $\{I_s(t), t \geq 0\}$, with finite state space $\{1, \ldots, m, m+1\}$, where $m+1$ is the absorption state. This distribution is characterised by the matrix representation (β, S) where β is the probability distribution of initiating the process from any state and S is the rate of transition among transient states. For more details on the PH distribution, we refer to He (2014), Chakravarthy and Neuts (2014).

The infinitesimal generator matrix of $I_s(t)$ in the context of EV charging can be represented by:

$$Q = \begin{pmatrix} S & S^0 \\ 0 & 0 \end{pmatrix}$$

where S (also called the *sub-generator matrix*), a square matrix of size m_s, captures the dynamics of the charging duration that, for example, can be attributed to the SoC levels in the EV being charged, and $\boldsymbol{S^0} = -Se$ representing the rate at which charging sessions are completed from various intermediate states. Throughout this study, e denotes a column vector of ones of appropriate size. β (also called the *sub-stochastic vector*) represents the probabilities of initiating the charging process from each state. These characteristics make the PH distribution particularly suitable for modelling the stochastic nature of EV charging durations, integrating variability and underlying Markovian dependencies into the queuing framework.

3.2. The queuing model

In our queuing model, EVs arrive at the station following an MAP characterised by the matrix pair (D_0, D_1). The average arrival rate of EVs under the MAP is defined as $\lambda = \theta D_1 e$, where θ satisfies the system of

equations $\theta(D_0 + D_1) = 0$ and $\theta e = 1$. The charging times of EVs follow a common PH distribution with PH representation (β, S). The average charging rate of EVs is given by $\mu = -1/(\beta S^{-1} e)$. The EV charging queue is illustrated in Figure 1.

The system is modelled as a continuous-time Markov chain (CTMC), described by the process

$$X(t) = \{N_q(t), I_a(t), I_{s1}(t), \ldots, I_{sc}(t), \quad t > 0\} \tag{1}$$

with state space: $S = \{\{0\} \times \{1, 2, \ldots, m_a\}\} \cup \{\{1, 2, \ldots\} \times \{1, 2, \ldots, m_a\} \times \{\{1, 2, \ldots, m_s\}^k\}\}$, where k denotes the number of busy connectors at time t. In this model, the states of the underlying MAP and PH processes, known as phases, combine to define the overall state of the CTMC. The state variables are defined as follows:

- $N_q(t)$: The total number of EVs in the station at time t, including those connected to chargers and those waiting in the queue ($N_q(t) = n$ for $n = 0, 1, 2, \ldots$).
- $I_a(t)$: The phase of the MAP governing the arrival process at time t.
- $I_{si}(t)$ for $i = 1, \ldots, c$: The phase of the PH distribution representing the charging time process at the ith connector.

The structure of the infinitesimal generator, Q, for this process is given by:

$$Q = \begin{array}{l} \text{Level } 0 \\ \text{Level } 1 \\ \vdots \\ \text{Level } c \\ \text{Level } (c+1) \\ \vdots \end{array} \begin{pmatrix} A_{0,0} & A_{0,1} & & & & & & \\ A_{1,0} & A_{1,1} & A_{1,2} & & & & & \\ & \ddots & \ddots & \ddots & & & & \\ & & & A_{c,c-1} & A_1 & A_0 & & \\ & & & & A_2 & A_1 & A_0 & \\ & & & & & \ddots & \ddots & \ddots \end{pmatrix} \tag{2}$$

The infinitesimal generator Q is structured into levels, where each level represents the total number of EVs in the station, including those being serviced and those waiting in the queue. The block matrices $A_{i,j}$ govern the transitions between these levels and are defined as follows:

- Level $n = 0$:
  - $A_{0,0} = D_0$: Represents internal transitions when there are no EVs in the system.
  - $A_{0,1} = D_1 \otimes \beta$: Governs transitions due to an arrival at level 0, when the system is empty. This accounts for the initiation of the charging process for the newly arrived EV.

- Level $0 < n \leq c$:

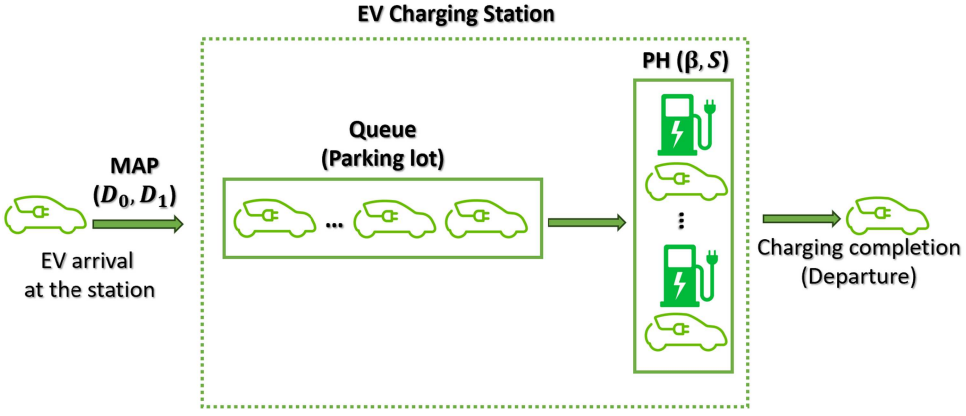


Figure 1. The structure of the proposed multi-server queuing model, by depicting the arrival, charging, and waiting processes within the station.

- $A_{n,n-1} = \mathrm{I}_{\mathrm{MAP}} \otimes (\sum_{\ell=1}^{n} \mathrm{I}_{\mathrm{PH}}^{\otimes(\ell-1)} \otimes \mathbf{S^0} \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes(n-\ell)})$: Governs charging completions that result in transitions from level n to $n-1$ EVs in the system.
- $A_{n,n+1} = D_1 \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes n} \otimes \beta$: Represents arrival transitions from level n to level $n+1$ EVs. Charging is initiated immediately, with no other transitions at the busy connectors.
- $A_{n,n} = D_0 \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes n} + \mathrm{I}_{\mathrm{MAP}} \otimes (\sum_{\ell=1}^{n} \mathrm{I}_{\mathrm{PH}}^{\otimes(\ell-1)} \otimes S \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes(n-\ell)})$: Describes internal transitions in the absence of arrivals or charging completions.

- Level $n > c$:
  - $A_2 = \mathrm{I}_{\mathrm{MAP}} \otimes (\sum_{\ell=1}^{c} \mathrm{I}_{\mathrm{PH}}^{\otimes(\ell-1)} \otimes (\mathbf{S^0} \otimes \beta) \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes(c-\ell)})$: Governs charging completions that result in transitions from level n to $n-1$ EVs in the system, with a queue present.
  - $A_0 = D_1 \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes c}$: Represents the arrival transitions from level n to $n+1$ EVs, where all the connectors are busy, and no charging begins.
  - $A_1 = D_0 \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes c} + \mathrm{I}_{\mathrm{MAP}} \otimes (\sum_{\ell=1}^{c} \mathrm{I}_{\mathrm{PH}}^{\otimes(\ell-1)} \otimes S \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes(c-\ell)})$: Describes internal transitions in the absence of arrivals or charging completions, with at least one EV waiting in the queue.

Thus, the process $X(t)$, represented in ((1)), is a level-dependent Quasi-Birth-and-Death (LDQBD) process, and the matrix geometric methods can be applied to study its stationary distribution.

3.3. Matrix geometric solution and performance measures

In Theorem 1, under the assumption of irreducibility for the underlying Markov chains, we prove that the LDQBD process $X(t)$ is ergodic, which implies the existence of a stationary distribution (He 2014). The stationary distribution of the process $X(t)$ is denoted by $\boldsymbol{\pi} = (\boldsymbol{\pi_0}, \boldsymbol{\pi}, \ldots)$, where:

- $\boldsymbol{\pi_0} = (\pi_{0,i})$ for $i = 1, \ldots, m_a$, describes the stationary distribution at level $n = 0$.
- $\boldsymbol{\pi_n} = (\pi_{n,i,j_1,\ldots,j_k})$ for $n > 0$, where $i = 1, \ldots, m_a$, $j_\ell = 1, \ldots, m_s$ for $\ell = 1, \ldots, k$, and $k = \min(n, c)$, represents the stationary distribution for levels $n > 0$.

The size of $\boldsymbol{\pi_n}$ is m_a at level $n = 0$, and $m_a \times (m_s)^k$ for levels $n > 0$. The stationary distribution $\boldsymbol{\pi}$ is the unique nonnegative solution to the system $\boldsymbol{\pi}Q = \mathbf{0}$, subject to the normalisation condition $\boldsymbol{\pi}e = 1$. The infinitesimal generator matrix Q is level-dependent up to level c and becomes level-independent beyond that. The expanded system of equations is given by:

$$\begin{aligned} &\boldsymbol{\pi_0} A_{0,0} + \boldsymbol{\pi_1} A_{1,0} = \mathbf{0}, \\ &\boldsymbol{\pi_{n-1}} A_{n-1,n} + \boldsymbol{\pi_n} A_{n,n} + \boldsymbol{\pi_{n+1}} A_{n+1,n} = \mathbf{0}, \quad \text{for } 0 < n \le c, \\ &\boldsymbol{\pi_{n-1}} A_0 + \boldsymbol{\pi_n} A_1 + \boldsymbol{\pi_{n+1}} A_2 = \mathbf{0}, \quad \text{for } n > c. \end{aligned} \tag{3}$$

For levels $n > c$, where Q becomes level independent, the matrix geometric method is applied to compute the rate matrix R, which satisfies the matrix equation $A_0 + RA_1 + R^2A_2 = \mathbf{0}$. For levels $n \le c$, the rate matrix $R(n)$, at each level, is computed iteratively using $A_{n-1,n} + R(n)A_{n,n} + R(n)R(n+1)A_{n+1,n} = 0$ for $1 \le n \le c$, where $R(c+1) = R$. Using this structure, $\boldsymbol{\pi_n}$ can be computed recursively by the formula ((4)).

$$\boldsymbol{\pi_n} = \begin{cases} \boldsymbol{\pi_{n-1}} R(n), & \text{for } 1 \le n \le c, \\ \boldsymbol{\pi_c} R^{n-c}, & \text{for } n \ge c+1. \end{cases} \tag{4}$$

To determine $\boldsymbol{\pi_0}$, the boundary equation $\boldsymbol{\pi_0} A_{0,0} + \boldsymbol{\pi_1} A_{1,0} = 0$ is used at level $n = 0$ subject to normalisation condition $\boldsymbol{\pi}e = 1$, expanded below:

$$\boldsymbol{\pi}e = \boldsymbol{\pi_0}\left(e + \sum_{k=1}^{c-1} R(1)R(2)\cdots R(k)e + R(1)\cdots R(c)[(I-R)^{-1}]e\right) = 1$$

To analyse the stability and efficiency of the queuing system, Theorem 1 provides the necessary and sufficient condition for the ergodicity of the process $X(t)$, under given irreducibility assumptions.

Theorem 1. (Ergodicity Theorem). *Assume that $D_0 + D_1$ and $S + \mathbf{S^0}\beta$ are irreducible. Then the process $X(t)$, described in (1), is ergodic (i.e. the queuing system is stable) if and only if $\rho = \lambda/\mu < 1$, where $\lambda = \boldsymbol{\theta} D_1 \mathbf{e}$ and $\mu = -1/(\beta S^{-1}\mathbf{e})$ are, respectively, the arrival and service rate of the system as defined earlier.*

Proof. In the process $X(t)$, since $\{I_a(t), t > 0\}$ and $\{I_{si}(t), t > 0\}$, for $i = 1, 2, \ldots, c$, are irreducible, all states communicate with each other. Next, we verify Neuts' condition for ergodicity (Neuts 1979), as also discussed in He (2014), Theorem 3.2.1. Let θ satisfy $\theta(D_0 + D_1) = 0$ and $\theta(S + \mathbf{S^0}\beta) = 0$, subject to the normalisation condition $\theta e = 1$, which represents the stationary row vector for the background states at levels $n > c$.

We first demonstrate that $\theta A_0 e = \lambda$, which corresponds to the long-run arrival rate derived from the MAP. Since e can be expressed as $\boldsymbol{e_{m_a}} \otimes e_{\mathbf{m_s^c}}$, where e_{m_a} and $e_{\mathbf{m_s^c}}$ are vectors of ones of size m_a and m_s^c, respectively. Thus, we obtain:

$$A_0\ e = (D_1 \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes c}) \quad (e_{m_a} \otimes e_{m_s^c}) = (D_1\, e_{m_a}) \otimes (\mathrm{I}_{\mathrm{PH}}^{\otimes c}\ e_{m_s^c}) = (D_1\, e_{m_a}) \otimes e_{m_s^c},$$

where $\mathrm{I}_{\mathrm{PH}}^{\otimes c}\ \ e_{m_s^c} = e_{m_s^c}$. Letting $\boldsymbol{\alpha} = D_1\ \ e_{m_a}$, with elements $\alpha_i = \sum_j (D_1)_{i,j}$ representing the total arrival transition rate from MAP phase i, we multiply $A_0 e$ by θ from the left. Expressing θ as the row vector $\theta_{i,j_1,\ldots,j_c}$ over the MAP phase i and PH phases $j_1, \ldots, j_c$, where $\boldsymbol{\theta_i} = \sum_{j_1,\ldots,j_c} \theta_{i,j_1,\ldots,j_c}$, we obtain:

$$\theta A_0\ e = \theta\left[\boldsymbol{\alpha} \otimes e_{m_s^c}\right] = \sum_{i=1}^{m_a} \sum_{j_1,\ldots,j_c} \theta_{i,j_1,\ldots,j_c} \quad \alpha_i = \sum_{i=1}^{m_a} \theta_i \alpha_i = \theta\boldsymbol{\alpha} = \theta(D_1 e_{ma}).$$

At the equilibrium level $n > c$, the underlying MAP chain is in its stationary phase distribution θ_i, subject to $\sum_i \boldsymbol{\theta_i} = 1$, thus yielding:

$$\theta A_0\ \ e = \theta D_1\ \ e_{m_a} = \lambda.$$

Next, we prove that $\theta A_2\, e = c\mu$. The block A_2 represents a single service completion by exactly one of the c independent busy connectors. For large $n > c$, any completion event at that level can be viewed as originating from one of the c identical PH processes:

$$A_2 e = \left(\mathrm{I}_{\mathrm{MAP}} \otimes \sum_{\ell=1}^{c} [\mathrm{I}_{\mathrm{PH}}^{\otimes(\ell-1)} \otimes (\mathbf{S}^0 \otimes \beta) \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes(c-\ell)}]\right)\left(e_{m_a} \otimes e_{m_s^c}\right)$$

Since $\mathrm{I}_{\mathrm{MAP}}\, e_{m_a} = e_{m_a}$, and using the identity $\mathrm{I}_{\mathrm{PH}}^{\otimes k}\, e_{m_s^k} = e_{m_s^k}$, each summand simplifies to:

$$\mathrm{I}_{\mathrm{PH}}^{\otimes(\ell-1)} e_{\mathbf{m_s^{\ell-1}}} = e_{\mathbf{m_s^{\ell-1}}}, \quad \mathrm{I}_{\mathrm{PH}}^{\otimes(c-\ell)} e_{\mathbf{m_s^{c-\ell}}} = e_{\mathbf{m_s^{\ell-1}}}$$

Thus, the term $(\mathbf{S^0} \otimes \beta) e_{\mathbf{m_s}}$ remains. Since each summand in

$$\sum_{\ell=1}^{c} [\mathrm{I}_{\mathrm{PH}}^{\otimes(\ell-1)} \otimes (\mathbf{S}^0 \otimes \beta) \otimes \mathrm{I}_{\mathrm{PH}}^{\otimes(c-\ell)}]$$

reflects the event that the ℓth connector completes its PH service, their sum accounts for the union of all such independent events. Consequently, the total completion rate from that level sums up to $c\mu$, leading to:

$$\mathbf{A}_2\ e\ \ =\ \ e_{m_a}\ \otimes\ (c\mu\ e_{m_s^c}).$$

Multiplying by θ from the left yields:

$$\theta\mathbf{A}_2\, e\ \ =\ \ \theta[e_{m_a} \otimes (c\mu\ e_{m_s^c})] = c\mu(\theta[e_{m_a} \otimes e_{m_s^c}]).$$

Since the entries of θ sum to 1, it follows that: $\sum_{i,j_1,\ldots,j_c} \theta_{i,j_1,\ldots,j_c}\ =\ 1$, which confirms $\theta\mathbf{A}_2\ \ e = c\mu$.

Thus, Neuts' condition for stability is satisfied if and only if:

$$\rho = \frac{\lambda}{c\mu} = \frac{\theta A_0 e}{\theta A_2 e} < 1 \;\rightarrow\; \theta A_0 e < \theta A_2 e \tag{5}$$

This completes the proof. □

To evaluate the performance of the queuing model, we focus on the following key performance metrics:

1. EV station utilisation factor (load factor):

$$\rho = (\theta D_1 e)(-\beta S^{-1} e)/c \tag{6}$$

2. The expected queue length $\mathbb{E}(L_q)$:

$$\mathbb{E}(L_q) = \sum_{n=2}^{\infty} (n-1)\,\boldsymbol{\pi}_n e = \boldsymbol{\pi}_1 R\,(I-R)^2 e \tag{7}$$

3. The expected waiting time in the queue $\mathbb{E}(W_q)$:

$$\mathbb{E}(W_q) = \frac{\mathbb{E}(L_q)}{\theta D_1 e} \tag{8}$$

Theorem 2 indicates the valid ranges that the elements of β may take to have a stable system.

Theorem 2. (Acceptable range of β). *For a queuing system with $m_s \geq 2$ phases in the charging process, the minimum value of β, required to ensure system stability is given by:*

$$\beta_1^{min} = \frac{(-c/\lambda) - \sum_{i=2}^{m_s} w_i (S^{-1}e)_i}{(S^{-1}e)_1 - \sum_{i=2}^{m_s} w_i (S^{-1}e)_i} \tag{9}$$

where:

- *c is the number of EV connectors,*
- *$\lambda = \boldsymbol{\theta} D_1 \boldsymbol{e}$ represents the arrival rate,*
- *$S^{-1}\boldsymbol{e}$ represents the vector obtained from inverse of rate matrix S of the charging process,*
- *$\sum_{i=2}^{m_s} w_i (S^{-1}\boldsymbol{e})_i$ is the weighted average for the elements of $S^{-1}\boldsymbol{e}$.*

Proof. To determine the minimum value of $\boldsymbol{\beta}$ that ensures the system remains stable, we must satisfy the condition $\rho < 1$. For β to be a valid probability vector, it must hold that $\sum_{i=1}^{m_s} \beta_i = 1$, for $\beta_i \geq 0$. By expressing the service rate as $\mu = \frac{-1}{\beta \cdot S^{-1}e}$, we adjust the first element of β, denoted as β_1, and distribute the remaining probability among the other $m_s - 1$ phases using the weights w_i:

$$\beta_i = (1-\beta_1) w_i, \;\; \text{for } i = 2, 3, \ldots, m_s, \;\; \text{subject to} \sum_{i=2}^{m_s} w_i = 1, \;\; \text{for} \;\; w_i \geq 0.$$

Next, the dot product $\beta \cdot S^{-1}e$ can be computed as follows:

$$\beta \cdot S^{-1}e = \beta_1 (S^{-1}e)_1 + \sum_{i=2}^{m_s} \beta_i (S^{-1}e)_i = \beta_1 (S^{-1}e)_1 + (1-\beta_1) \sum_{i=2}^{m_s} w_i (S^{-1}e)_i$$

To ensure $\rho < 1$, we set the minimum service rate as $\mu^{\min} = \frac{\lambda}{c}$. Substituting this into the utilisation expression:

$$-1 = \frac{\lambda}{c}(\beta^{\min} \cdot S^{-1}e) = \frac{\lambda}{c}\left[\beta_1^{\min} (S^{-1}e)_1 + (1-\beta_1^{\min}) \sum_{i=2}^{m_s} w_i (S^{-1}e)_i\right]$$

which yields:

$$\frac{-c}{\lambda} = \beta_1^{\min}[(S^{-1}e)_1 - \sum_{i=2}^{m_s} w_i (S^{-1}e)_i] + \sum_{i=2}^{m_s} w_i (S^{-1}e)_i$$

Thus, the minimum value of β_1 required to satisfy the stability condition $\rho < 1$ is given by (9). This finalises the proof.

3.4. Computational complexity analysis of the MAP/PH/c framework

The computational complexity of the proposed MAP/PH/c framework is driven primarily by the dimension of the state space associated with each level of the QBD representation. At any level with k busy servers, the system is fully described by the MAP phase together with the joint service-phase configuration of those k servers. Because the servers are indistinguishable, the service state depends only on how many of the k servers occupy each of the m_s service phases. The number of such occupancy patterns is the number of weak compositions of k into m_s parts, given by $\binom{k + m_s - 1}{m_s - 1}$ (Neuts 1979; Chakravarthy and Alfa, 1996). Summing over all possible values of $k = 0, \ldots, c$ and multiplying by the m_a number of MAP phases yields the block dimension:

$$d(m_a, m_s, c) = m_a \sum_{k=0}^{c} \binom{\mathrm{k} + \mathrm{m_s} - 1}{\mathrm{m_s} - 1} = m_a \binom{c + m_s}{m_s},$$

which is the standard combinatorial expression governing the size of the QBD blocks in MAP/PH/c queues.

The dominant matrix-analytic computations are carried out on dense $d(m_a, m_s, c) \times d(m_a, m_s, c)$ matrices. As a consequence, the computational cost grows as $O(d(m_a, m_s, c)^3)$, with memory requirements growing as $O(d(m_a, m_s, c)^2)$. Since the truncation level affects only the number of QBD levels and not the dimension of each block, its influence on runtime is relatively minor. In practice, the phase-space dimension $d(m_a, m_s, c)$ is therefore the key determinant of computational effort. To keep the model computationally manageable while still capturing essential behavioural variability, we select m_a and m_s via empirical fitting and clustering, thereby avoiding unnecessary inflation of the phase structure.

It is important to note that prior studies employing MAP/PH queuing models rarely provide such a detailed examination of computational tractability. Thus, this work fills an important gap by explicitly quantifying the computational costs and demonstrating that practical implementation is feasible for real charging-station settings, with clear avenues for further scaling and optimisation. The measured runtimes and empirical complexity assessment for the fitted model are reported in Section 4, confirming the practical implementation of the proposed approach.

4. Model validation and case study

4.1. Numerical example

In this section, we evaluate the performance of the developed MAP/PH/c queuing model in the context of an EV charging station. We begin with a numerical example of a charging station with one to three connectors, where EVs arrive following a MAP process and charging durations are modelled by a PH distribution. While the example focuses on a limited number of connectors, our model is capable of handling a larger number of connectors, depending on the structure of the arrival and service processes, and this does not pose a limitation in our study. Key performance metrics, such as the average queue length (L_q) and the average waiting time (W_q), are analysed under the conditions of correlated arrivals and variable service times. To ensure the reliability of the results, we validate these performance metrics by comparing the MAP/PH/c model outcomes to those of the well-known M/M/c queuing model. This comparison confirms the consistency and robustness of our developed model, a principle also reflected in Asmussen and Møller (2001), Chakravarthy (2009), where a similar performance comparison was conducted. The results illustrate the model's ability to capture key stochastic behaviours, such as correlated

arrival patterns and variability in the charging process, that are characteristic of modern EV charging stations.

We consider an EV charging station where vehicles arrive according to a correlated process, modelled by a second-order MAP. In this example, the arrival process exhibits a negative correlation, represented by the following matrices:

$$D_0 = \begin{bmatrix} -2 & 1 \\ 0 & -1.5 \end{bmatrix}, \quad D_1 = \begin{bmatrix} 1 & 0 \\ 1 & 0.5 \end{bmatrix}, \tag{10}$$

Here, the negative correlation implies that following a cluster of vehicle arrivals, there is a tendency for fewer new arrivals to follow immediately. This behaviour can occur after the end of a busy commute, when traffic gradually decreases, or when drivers see a crowded station with long queues and choose not to stop. In both scenarios, an initial wave of arrivals is often followed by a quieter period before the next cluster begins. The charging session of each vehicle is also described by a PH distribution of order two with:

$$\beta = (0.8,\ 0.2), \quad S = \begin{bmatrix} -5 & 0 \\ 0.5 & -1 \end{bmatrix} \tag{11}$$

In this example, the parameters of the MAP indicate an effective arrival rate of $\lambda = 1.25$ vehicles per hour, corresponding to roughly one to two vehicles arriving each hour, a realistic scenario for a small local EV charging station. The PH service distribution yields a mean charging rate of $\mu = 2.63$ vehicles per hour with an average charging duration of $1/\mu = 0.38$ hours (approximately 23 minutes), which is consistent with typical DC fast-charging behaviour at moderate SoC levels. Together, these values represent a moderate traffic intensity and provide a realistic test setting for evaluating the performance of an EV charging station under the proposed MAP/PH/c queuing model.

To validate our MAP/PH/c model and its implementation, we analysed how key performance metrics, specifically average waiting time W_q (per hour) and average queue length L_q (in vehicles), vary with changes in the utilisation factor. The consistency observed in our results aligns with established findings in the literature, further underscoring the accuracy and robustness of our proposed model (Chakravarthy 2009; Dudina, Kim, and Dudin 2013). As shown in Figure 2, both W_q and L_q increase with higher utilisation, a behaviour consistent with fundamental principles of queuing theory. This validation not only demonstrates the feasibility of our developed model but also highlights its reliability to accurately capture system dynamics under varying traffic intensities.

Table 3 provides a comparison between the MAP/PH/c queue results and the M/M/c queue. The upper part of Table 3 presents the results of the M/M/c queue, which serves as a baseline for comparison. The middle part demonstrates the results of a simplified version of the MAP/PH/c model, where the MAP reduces to a Poisson process and the PH service time is an exponential distribution. In this representation, correlations in the arrival and charging time processes are ignored, and the model aligns closely with the M/M/c queue. The metrics under different connector configurations confirm that the simplified MAP/PH/c model reproduces the expected behaviour of the M/M/c queue, which validates the accuracy of our model. The lower part of Table 3 provides results for the MAP and PH representations provided in Equations (10) and (11). The results highlight that by incorporating variability and correlation in arrival and charging time processes, our model predicts the performance of the EV charging system more accurately. The MAP/PH/c model represents more realistic estimates for L_q and W_q while maintaining the same idle probability ($\boldsymbol{\pi_0}$) and utilisation ($\boldsymbol{\rho}$) as the M/M/c model.

Figure 2 compares the steady-state expected queue length (vehicles) and waiting time (hours) for the M/M/c and MAP/PH/c models under connector configurations $c = 1, 2, 3$. For all cases, the MAP/PH/c model and the M/M/c model produce similar results with low utilisation factors, indicating consistent behaviour when congestion effects are minimal. As the utilisation factor increases, the MAP/PH/c model predicts larger values of L_q and W_q compared to the M/M/c model, demonstrating its ability to account for complexities in the arrival and service time processes. In practice, increasing the number of connectors creates a larger system where correlated arrivals and service time variability have a more significant cumulative impact on the system congestion. The MAP/PH/c model can effectively capture these dynamics, while the M/M/c model underestimates the actual congestion in the system. This confirms that the proposed MAP/PH/c model is a robust queuing framework capable of representing the dynamics and complexities inherent in EV charging stations.

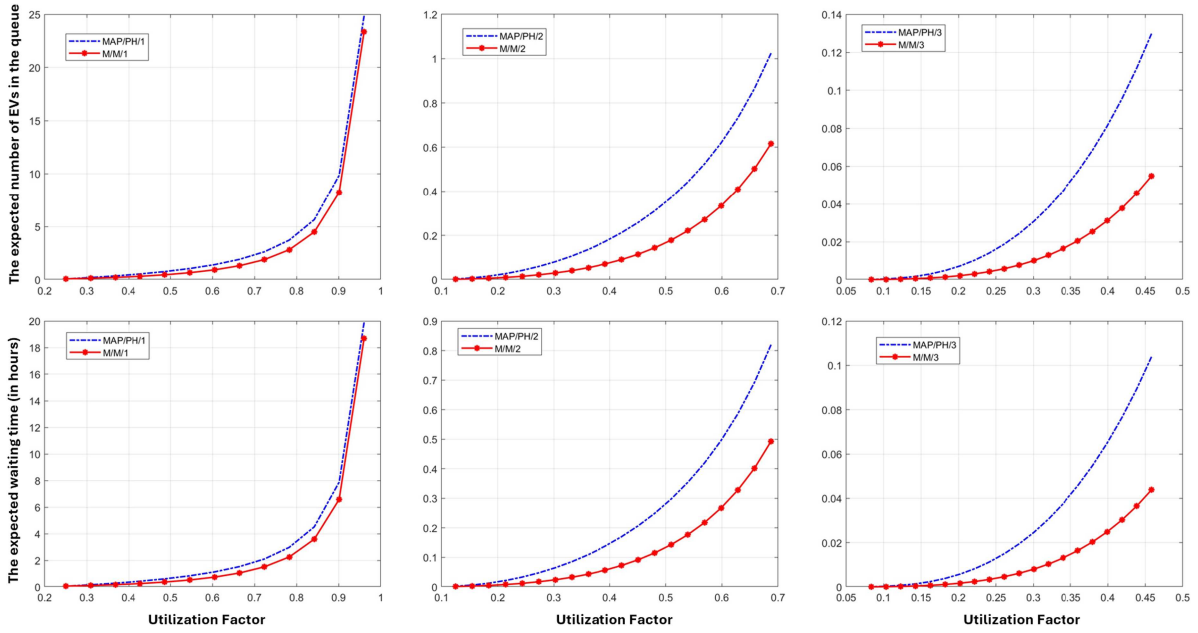


Figure 2. Comparison of the key performance metrics for the MAP/PH/c and M/M/c models. This figure illustrates how the average queue length and waiting time vary for the MAP/PH/c and M/M/c models across different server configurations.

Table 3. Comparison of M/M/c and MAP/PH/c models for $c = 1$ to $c = 5$.

Model	Rates	Number of connectors (c)	π_0	ρ	L_q	W_q
M/M/c		1	0.5247	0.4753	0.4305	0.3444
	$\lambda = 1.25$, $\mu = 2.63$	2	0.6160	0.2376	0.0142	0.0114
		3	0.6213	0.1584	0.0008	0.0007
		4	0.6217	0.1188	0.0001	0.0000
		5	0.6217	0.0951	0.0000	0.0000
MAP/PH/c (1st)	MAP: $D_0 = -1.25$, $D_1 = 1.25$	1	0.5247	0.4753	0.4305	0.3444
	PH: $S = -2.63$, $\beta = 1$	2	0.6160	0.2376	0.0284	0.0228
		3	0.6213	0.1584	0.0025	0.0020
		4	0.6217	0.1188	0.0002	0.0002
		5	0.6217	0.0951	0.0000	0.0000
MAP/PH/c (2nd)	MAP: Equation (10)	1	0.5250	0.4841	0.7162	0.5730
	PH: Equation (11)	2	0.6142	0.2604	0.0387	0.0310
		3	0.6178	0.1692	0.0027	0.0022
		4	0.6180	0.1221	0.0002	0.0001
		5	0.6180	0.0944	0.0000	0.0000

The table compares key performance metrics (steady-state idle probability, utilisation, queue length, and waiting time) across M/M/c and MAP/PH/c models under various configurations of charging connectors.

4.2 Case study: EV charging data

In this section, we detail a real-world application of the proposed MAP/PH/c framework using a dataset collected from EV users in Saskatchewan, Canada. The selected dataset includes charging sessions from both public and private charging locations. The Start and End SoC values for each dataset were automatically recorded by the telemetry device installed on the vehicle, which logs the SoC at the time the vehicle is plugged into the charger and again when it is unplugged. Approximately 53% of the vehicles in the dataset where Tesla models, with the remaining 47% coming from 14 other manufacturers. Due to the dominance of Teslas in our dataset, and especially because Tesla is one of the few networks with publicly documented charging-rate behaviour at different SoC levels, we model a Tesla Supercharger station. According to Tesla's official website (Tesla 2025), Tesla Superchargers can charge a Tesla EV from 20% to 80% in approximately 15 to 20 minutes, at which point the system automatically stops charging at 80% to reduce congestion at high-use stations, especially on highways. Tesla users can choose to charge beyond 80%, but the charging rate slows dramatically beyond this point, with a full charge (100%) taking up to an hour. To reflect this

behaviour, we excluded charging sessions lasting more than 50 minutes to incorporate all types of EVs (i.e. Tesla or non-Tesla) using the Tesla supercharger station. These parameters were selected to ensure that the modelled system aligns with the operational practices of Tesla superchargers and captures the dominant charging behaviour observed in the dataset. Nonetheless, our model can be utilised effectively for other DCFC charging stations.

Furthermore, a time-series analysis of the dataset (one year) revealed significant temporal correlations in the arrival patterns of EVs at charging stations. As shown in Figure 3, the dataset shows periods of high and low activity aligning with user behaviour (e.g. peak commute times coinciding with busy periods). These temporal correlations highlight the need for a queuing framework such as MAP/PH/c, which can accurately capture the time-dependent and correlated nature of EV arrivals and, therefore, the real-world usage dynamics at any DCFC charging station.

To evaluate the developed MAP/PH/c queuing system's performance in a real-world scenario, we fitted an MAP to the inter-arrival times of the charging sessions and a Phase-Type (PH) distribution to their corresponding charging durations. All fitting procedures were conducted in MATLAB using the BuTools package (Horvath and Telek 2017).

$$D_0 = \begin{bmatrix} -0.0505 & 0 & 0 \\ 0 & -0.0781 & 0 \\ 0 & 0 & -0.1178 \end{bmatrix} \quad D_1 = \begin{bmatrix} 0.0486 & 0.0018 & 0.0000 \\ 0.0079 & 0.0520 & 0.0182 \\ 0.0001 & 0.0135 & 0.1042 \end{bmatrix} \tag{12}$$

The fitted MAP representation of inter-arrival times, shown in Equation (12), models the arrival process using three phases: low, moderate, and high Demand periods, which align with the temporal variations observed in Figure 3. Low demand occurs during the early morning and late-night hours, moderate demand is observed mid-day, and high demand coincides with peak commute hours (6 AM to 9 AM and 4 PM to 7 PM).

The number of phases and their interpretation were guided by descriptive analysis of the empirical inter-arrival times and direct inspection of the time series pattern, as this ensured that the MAP structure reflected the natural fluctuations present in the data. Rather than batching or discretizing arrival times into time blocks, our modelling approach preserves the continuity of the inter-arrival process, which allows the MAP to retain the full temporal resolution and correlation structure of the observed data.

The off-diagonal elements in D_0 are zero, as transitions between demand periods cannot occur without an arrival. Conversely, the off-diagonal elements of D_1 capture transitions between phases that result in an arrival. For example, the transition rate from low demand to high demand, or vice versa, is nearly zero, indicating that abrupt shifts between these phases are highly unlikely. Instead, transitions occur progressively, for example, from low to moderate demand or moderate to high demand. The higher values in the

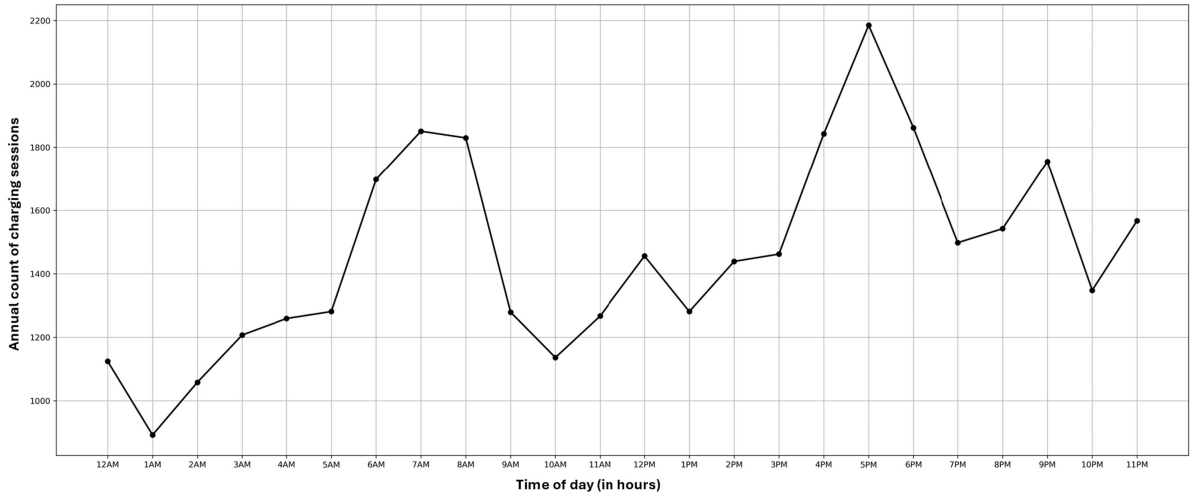


Figure 3. Temporal pattern of EV arrivals over one year, illustrating fluctuations over 24 hours and highlighting varying demand periods.

diagonal elements of D_1 demonstrate that arrivals within each phase are highly probable, implying strong correlations in arrivals within the same time window. This result highlights the MAP's ability to successfully capture positively correlated arrival patterns and gradual temporal variations.

To characterise the relationship between SoC and charging duration, and to inform the structure of the PH distribution, we conducted a detailed exploratory analysis of the empirical charging data. Because SoC is only observed at the start and end of each session, these two values, along with the observed charging time and the effective SoC increment ('SoC Jump'), were used to quantify charging patterns across the dataset. These variables were then standardised and subjected to K-means and hierarchical agglomerative clustering. As shown in Figures 4 and 5, the hierarchical dendrogram and the K-means elbow plot consistently indicate the presence of four natural groups of charging sessions. The vertical axes in Figures 4 and 5, respectively, represent the complete-linkage distance defined as the maximum pairwise dissimilarity between points in the clusters being merged and the total within-cluster sum of squared distances (inertia), which measures how compact the clusters are.

The four identified groups correspond to distinct SoC regimes that align with known fast-charging behaviour: sessions beginning below 20%, sessions beginning between 20–70%, sessions beginning around 70–80%, and sessions beginning around 80% SoC. For each cluster, the representative Start SoC and End SoC values were defined as the cluster means, while the representative charging duration was defined as the median to mitigate the influence of skewness. The proportional weight of each group reflects the number of observed sessions falling into that cluster. The cluster characteristics and internal validation metrics are summarised in Table 4. As can be seen, the Davies-Bouldin indices (all below 1.1), Silhouette scores (0.42–0.44), and Calinski-Harabasz values (exceeding 200) collectively confirm the robustness of a four-cluster partition. These data-driven clusters directly informed the selection of a four-phase PH service-time distribution. The PH-fitting algorithm was initialised using these groups to ensure that both the number of phases and the interpretation of estimated transition rates are grounded in empirical charging behaviour.

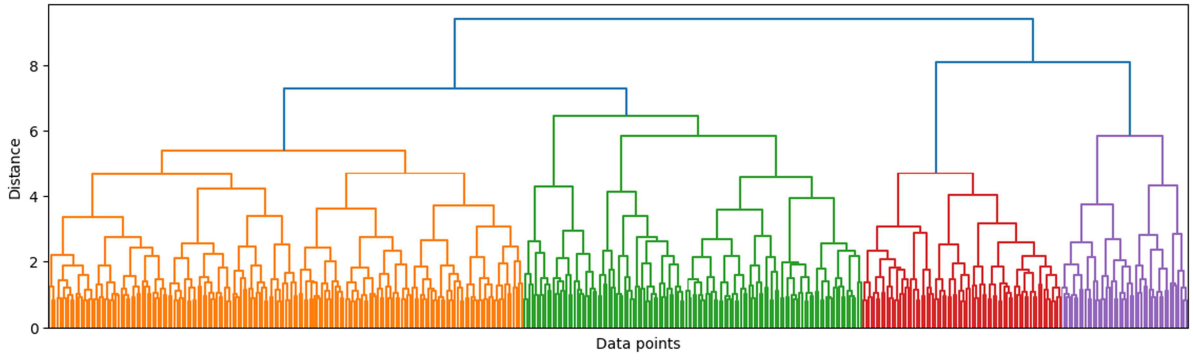


Figure 4. Hierarchical clustering dendrogram using complete linkage and truncated with lastp ($p = 430$).

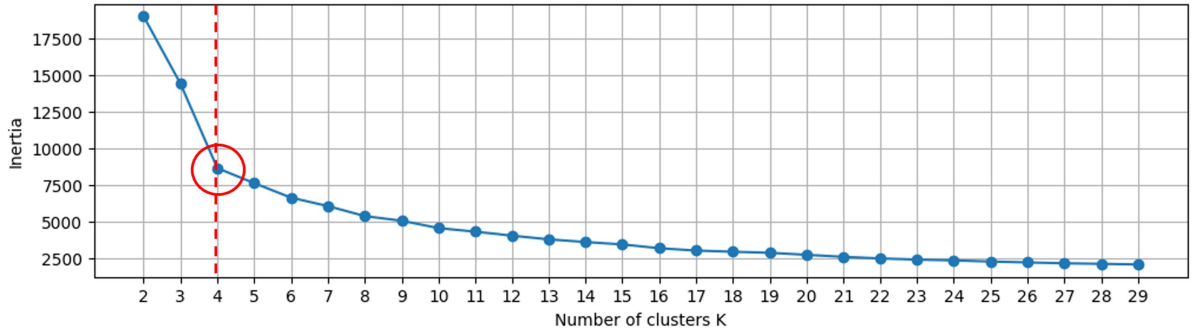


Figure 5. Elbow plot for K-means clustering. The elbow indicates the presence of 4 clusters.

Table 4. Consolidated clustering results: K-means clusters, hierarchical clusters, and internal validation metrics.

Method/Cluster	Start SoC	End SoC	SoC jump	Charging time (min)	Proportion
K-means Clusters					
KM–0	16.2538	53.7085	37.4547	19.967	0.3429
KM–1	26.6826	72.8838	46.2012	24.184	0.5421
KM–2	73.7766	82.3566	8.5800	18.217	0.1091
KM–3	80.4038	82.5876	2.1837	11.384	0.0060
Hierarchical Clusters					
HC–0	18.0832	58.9803	40.8972	24.734	0.3542
HC–1	24.0763	73.0678	48.9915	32.850	0.5942
HC–2	75.2754	80.5454	5.2701	20.317	0.0491
HC–3	79.5564	83.6338	4.0774	12.860	0.0025
Validation Metrics					
Davies–Bouldin Index	KM: 0.8276			HC: 1.0183	
Silhouette Score	KM: 0.4177			HC: 0.4408	
Calinski–Harabasz	KM: 405.16			HC: 219.50	

$$\beta = [0.4070,\ 0.5925,\ 0.0005,\ 0] \quad S = \begin{bmatrix} -0.3851 & 0.0007 & 0 & 0 \\ 0 & -0.1377 & 0.1377 & 0 \\ 0 & 0 & -0.1377 & 0.1377 \\ 0 & 0 & 0 & -0.1377 \end{bmatrix} \tag{13}$$

The fitted PH representation of charging durations in Equation (13) uses a four-phase structure that was determined through a data-driven clustering analysis of the observed charging sessions: Almost Empty (less than 20%), Moderately Charged (20% to 70%), Nearly Charged (70% to 80%), and Sufficiently Charged (around 80%). The *Fully Charged* state at 100% SoC is treated as the absorbing state of the PH distribution. This data-informed phase structure ensures that the PH model aligns with observed charging behaviour and captures the underlying physical characteristics of the charging process.

The elements of S define the transition rates between the SoC phases, capturing the natural progression of the charge, with zero transition rates from higher SoC levels to lower SoC levels. The consistent off-diagonal transition rates (0.1377) across the second and third phases reflect the Tesla Superchargers' ability to maintain a steady charging speed from 20% to 80% SoC, with the lower transition rates from Almost Empty to Moderately Charged and a slower charging speed when the battery is almost depleted. The initial probability vector, $\beta = [0.4070, 0.5925, 0.0005, 0]$, indicates that most vehicles start charging with SoC levels below 70%, reflecting users' preference to avoid charging at Sufficient SoC levels to prevent battery degradation. The higher probability of starting at Moderately Charged compared to Almost Empty aligns with real-world behaviour, suggesting that most users charge up before their SoC drops to critically low levels to mitigate range anxiety. This PH representation effectively captures the variability in charging durations and aligns with the practical charging dynamics observed at Tesla superchargers.

Figure 6 depicts the joint density of SoC and charging duration for the four SoC intervals used in the PH representation. Here, the charging time refers to the total duration required to complete the session, starting from the initial SoC. Thus, the End SoC of a session may fall outside the starting SoC interval shown in each subplot. The density plots clearly reveal distinct charging-time patterns across the four SoC ranges. These distributional differences directly align with the PH-fitted transition rates and support the data-driven decision to structure the PH distribution into four SoC-based phases. This structured approach provides a quantitative, SoC-aware foundation for constructing the PH distribution, as well as a clear pathway for future extensions, such as estimating separate PH models conditioned on SoC intervals or allowing service rates to vary dynamically with SoC during charging.

The arrival and charging rates at the station, derived from the MAP and PH representations, were 0.068 and 0.0717 per minute, respectively (Table 5). Table 5 also presents a comparative analysis of key statistical metrics, such as mean, variance, and skewness, for both the empirical data and the fitted MAP and PH distributions. The close alignment of these moments indicates that the proposed models accurately reproduce the variability observed in real charging sessions. To further assess distributional accuracy, several goodness-of-fit measures that capture the deviation between the empirical and fitted distributions are reported, namely, the KS statistic, RMSE, and MAPE. These measures function in a complementary manner, as the KS statistic captures the maximum vertical distance between the cumulative distribution functions, while RMSE and MAPE describe the differences between discretized probability masses to determine how well the fitted distributions reproduce the central behaviour and tails of the empirical

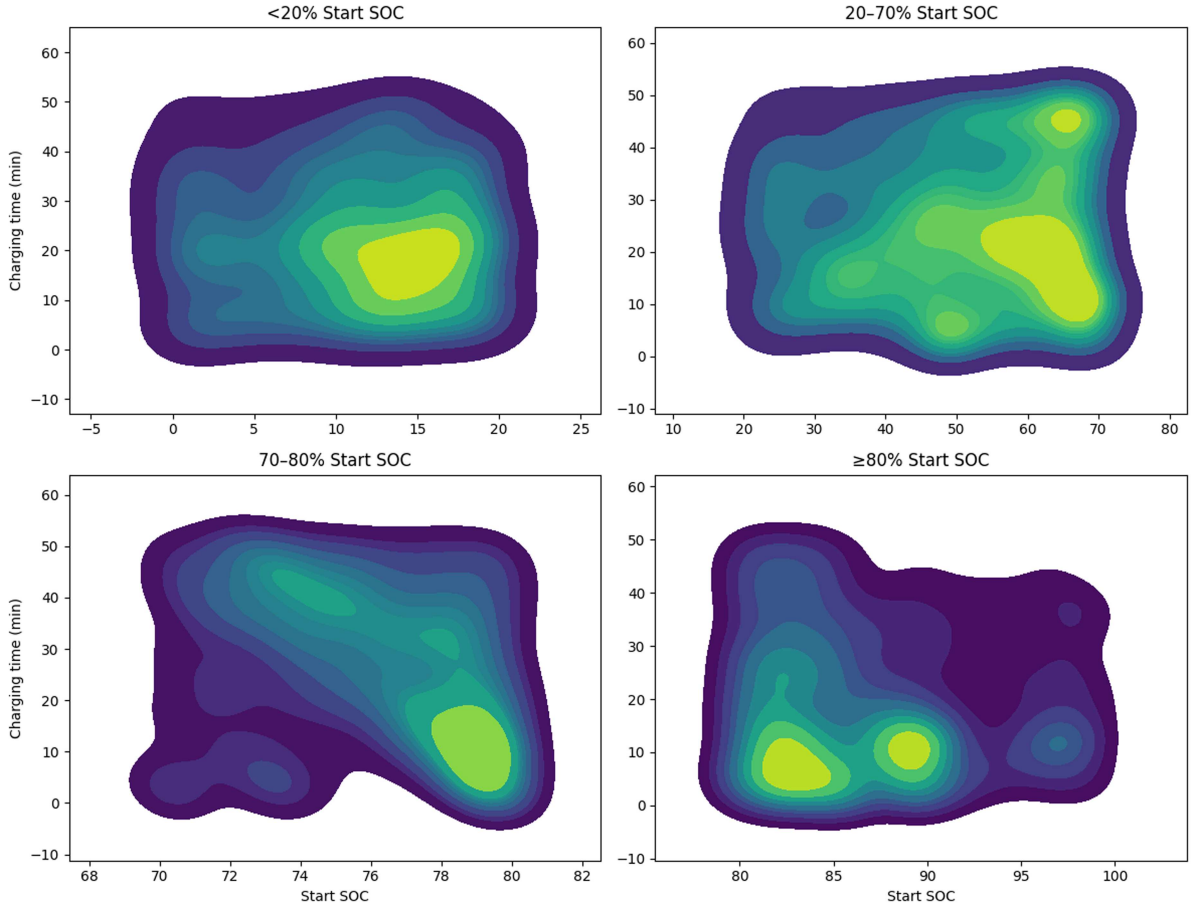


Figure 6. KDE of charging time distributions across SoC phases.

Table 5. Comparison of empirical and fitted results for the arrival (MAP) and service (PH) processes, including distributional goodness-of-fit metrics.

	Arrival process (MAP)		Service process (PH)	
	Empirical	Fitted MAP	Empirical	Fitted PH
Arrival Rate	0.06871	0.06803	–	–
Service Rate	–	–	0.07166	0.07149
Moments:				
Mean	14.7157	14.5553	13.9543	13.9887
Variance	269.1516	268.0217	184.7421	185.4884
Skewness	2.5695	2.5909	0.98518	0.9582
Goodness-of-Fit:				
KS Statistic		0.0133		0.0557
RMSE		0.0018		0.0093
MAPE		0.0395 (3.95%)		0.0865 (8.65%)

This table provides a comparative analysis of key statistical measures for the empirical and fitted arrival and service processes. The results confirm the accuracy of the fitted MAP and PH distributions in capturing the observed data characteristics.

distributions. The consistently low values of these metrics indicate that the fitted models closely approximate the shape and dispersion of the empirical distributions. Together, these results confirm the fitted MAP and PH representations' robust ability to capture real EV charging behaviour.

Figures 7 and 8 illustrate the fitting results, which show the MAP and PH distributions' ability to effectively capture the arrival and service time patterns of EV charging sessions. Figure 7 highlights the performance of the fitted MAP in modelling both the distribution and temporal correlation structure of inter-arrival times. As shown in Panel (a), the fitted MAP accurately captures the positive correlation

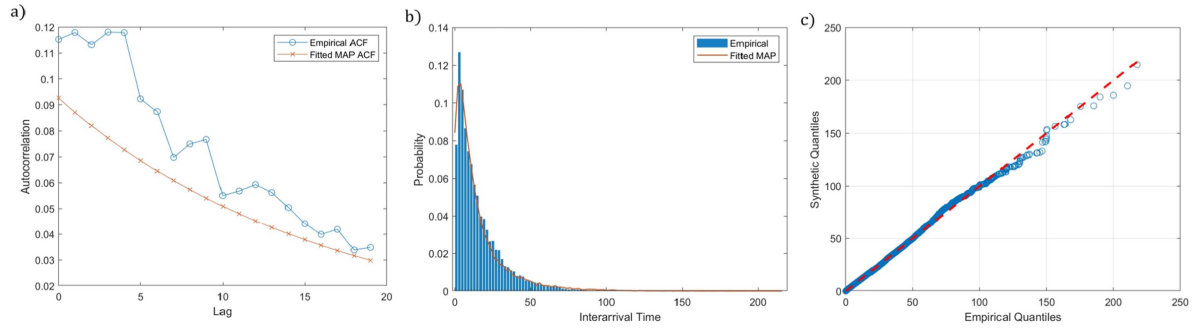


Figure 7. Panels a, b, and c illustrate the fitted MAP's effectiveness in capturing the inter-arrival time distribution and correlation patterns of EV charging sessions, confirming the model's ability to represent real-world arrival dynamics accurately.

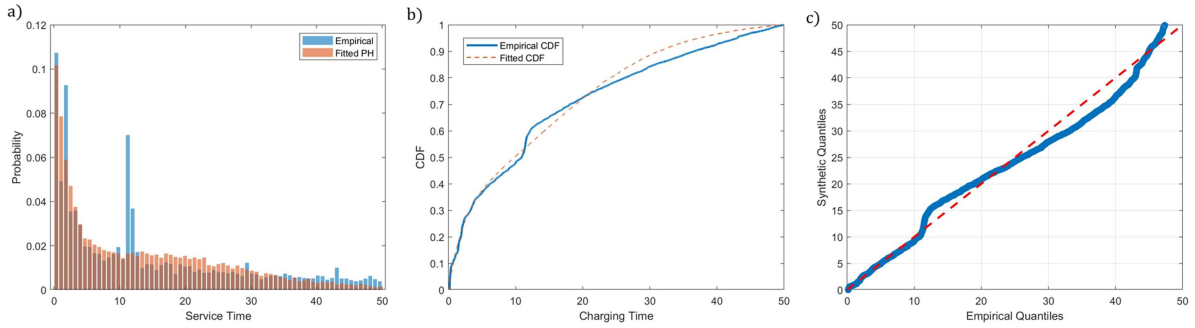


Figure 8. Panels a, b, and c depict the fitted PH distribution's ability to capture the variability in EV charging durations accurately, thus providing a realistic representation of the observed charging behaviour.

between arrival patterns, which aligns with the descriptive analysis of our dataset, wherein we observed multiple vehicles often arriving in clusters during specific time windows. Panel (b) presents a comparison between the empirical histogram of EV inter-arrival times and the fitted MAP. As the result show, the model closely approximates the probability distribution of arrival occurrences. Finally, the Q-Q plot in Panel (c) confirms the model's ability to replicate the cumulative distribution of inter-arrival times with minimal deviations, even at the distribution tails, further validating the effectiveness of the fitted MAP.

Similarly, Figure 8 illustrates the effectiveness of the PH distribution in capturing the variability of EV charging durations. Panel (a) depicts the empirical histogram of EV charging durations-filtered to include only sessions lasting less than 50 minutes-alongside the fitted PH distribution. A noticeable spike around the 12–13 minutes charging time can be attributed to the operational behaviour of Tesla EVs, which often terminate charging automatically at approximately 80% SoC. Notably, the data collection system records each charging session separately, which means that any continuation of charging after disconnection is logged as a new session rather than an extension of the original one. Due to this data tracking limitation, the empirical distribution contains a high concentration of sessions ending at 80%. However, the fitted PH distribution, which is the distribution of time until reaching a Fully Charged state (100% SoC), smooths this behaviour across a broader range, reflecting the general charging time patterns observed at Tesla Supercharger stations. This behaviour was not observed among all EVs in the dataset; rather, it was predominant among Teslas, which accounted for the majority of the dataset.

The comparison of empirical and fitted cumulative distribution functions (CDFs) shown in Panel (b) establishes the fitted PH model's ability to capture the overall distribution of charging times effectively. However, the slight deviations between the two curves suggest localised variations in the empirical data, possibly due to behaviour and charging habits. This pattern is further supported by the Q-Q plot in Panel (c), which shows a strong correlation between the empirical data and the fitted PH distribution, with minor deviations observed primarily at the distribution tails.

Overall, the fitting results confirm that the MAP and PH distributions successfully capture the statistical properties of EV inter-arrival times and charging durations. These models provide a robust foundation for analysing and optimising EV charging station operations, particularly for Tesla Supercharger networks, where understanding such patterns is critical for efficient station management and planning.

To utilise our MAP/PH/c queuing model for Tesla Supercharger stations, we calculated its stationary distribution using the structure and methodology outlined in Section 3. Figure 9 illustrates the relationship between the charging rate (μ), the number of connectors (c), and the key performance metrics. Panel (a) shows that the utilisation factor (ρ) decreases as the charging rate increases, with this trend being more prominent in systems with a higher number of connectors. As can be seen, at a current arrival and charging rate of 0.068 and 0.071 per minute, respectively, the station utilisation is expected to peak at approximately 85% with one connector, indicating a potential risk of overloading in this configuration. However, increasing the number of connectors and improving their charging rates causes the utilisation factor to steadily decline, thus confirming the station's stability. This finding serves as strong evidence of our model's capacity to accurately capture queuing behaviour under the fitted arrival and service processes. Furthermore, Panels (b) and (c) depict the trend towards declining expected queue lengths and waiting times, respectively, as the charging rate increases. As shown in Panel (b), congestion can be effectively mitigated by installing more than two connectors, as this would reduce the average queue length from nearly 30 EVs to fewer than six per hour. Similarly, Panel (c) indicates that the average wait time decreases from approximately 8 minutes to less than 1 minute under the current utilisation of the station.

Due to the characteristics of the utilised dataset and the current system arrival and service rates, the station is expected to explode when operating with a single connector (Figure 10). This issue gradually

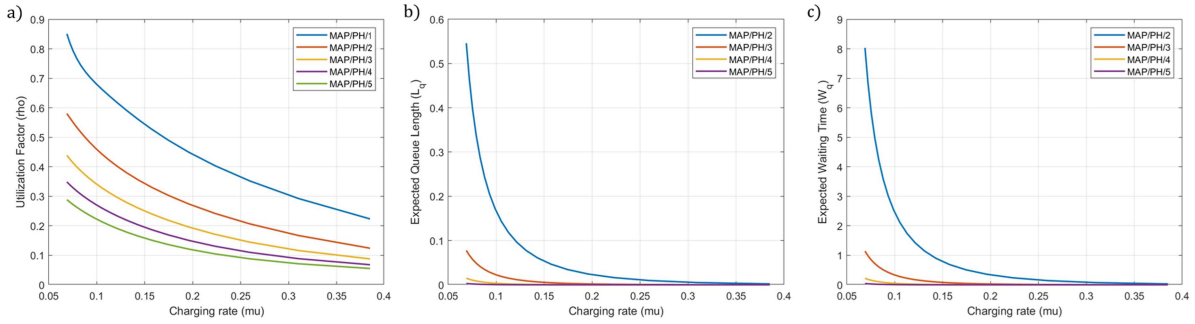


Figure 9. Key performance metrics obtained from the MAP/PH/c model using the fitted processes. Panels a, b, and c show how varying the number of connectors and charging rates impacts key performance metrics such as utilisation, expected queue length, and waiting time.

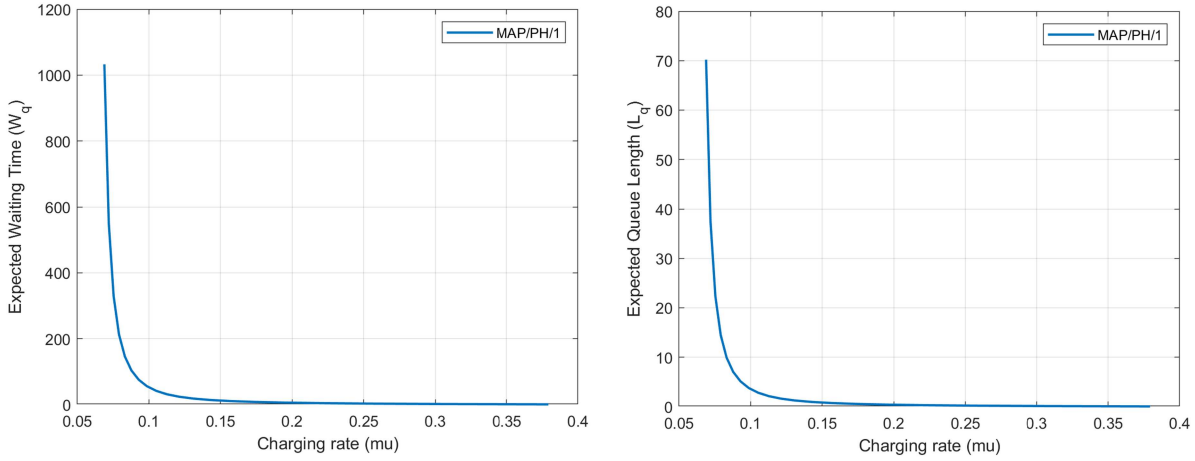


Figure 10. The impact of a single charging connector on station utilisation, queue length, and waiting times, emphasising the station's limitations under current conditions.

improves as the connector charging rate is increased, with optimal performance expected to be achieved at three times the existing charging rates. However, such a charging rate has not yet been achieved with current charging technologies. Therefore, the one-connector configuration was excluded from further analysis in Figure 9, as we wanted to focus on evaluating the performance measures of more practical, multi-connector configurations.

Table 6 presents the key performance measures calculated for the observed utilisation based on current arrival and charging rates in systems with different numbers of connectors. We also evaluated the Erlang C formula, or the probability of queue formation (denoted by P_c), as this metric reflects the likelihood that all available connectors are occupied when a new EV arrives:

$$P_c = P(N(t) > c) = \sum_{n=c+1} \sum_{i=1}^{m_a} \sum_{j_1=1}^{m_s} \cdots \sum_{j_k=1}^{m_s} \pi_{n,i,j_1,\cdots,j_k}$$

The remaining performance metrics were calculated based on the formulations outlined in Section 3.3. Table 6 compares the performance metrics of the M/M/c and fitted MAP/PH/c models across systems with different numbers of connectors. Both models show an expected reduction in congestion as capacity increases. However, the MAP/PH/c model consistently yielded higher queue formation probabilities, queue lengths, and waiting times across all configurations. This reflects the additional variability and correlation present in the empirical arrival and charging time data. Conversely, the M/M/c model systematically underestimated congestion, especially at lower capacities. Although the gap between the two models narrows as the number of connectors increases, the MAP/PH/c results remain more conservative and better aligned with the stochastic characteristics observed in real charging behaviour.

Table 6 also comprehensively shows how increasing the number of connectors impacts various performance indicators. The driven results from the MAP/PH/c model align with those presented in Figures 9 and 10, which reinforces the premise that adding more connectors can effectively mitigate congestion at the station. The results presented in Table 6 clearly show that the probability of queue formation decreases significantly as the number of connectors increases, from almost 82% to 18%, thus ensuring better service for a greater number of EVs without risking system overload.

Table 7 summarises the theoretical block dimensions and empirical runtimes for our proposed model with $m_a = 3$ MAP phases and $m_s = 4$ PH phases, which were evaluated on the reference machine (Intel Core Ultra 7 155 H, 1.40 GHz, 32 GB RAM).

Table 6. Comparison of queueing performance metrics for M/M/c and fitted MAP/PH/c models across different numbers of connectors.

Model	Rates	Number of connectors (c)	π_0	P_c	ρ	L_q	W_q
M/M/c		1	0.0515	0.7997	0.9485	17.4845	257.2199
	$\lambda = 0.0683$, $\mu = 0.0717$	2	0.3566	0.1452	0.4743	0.2751	4.0460
		3	0.3837	0.0252	0.3162	0.0369	0.5424
		4	0.3869	0.0041	0.2371	0.0053	0.0781
		5	0.3873	0.0006	0.1897	0.0007	0.0105
MAP/PH/c	MAP: Equation (11)	1	0.0502	0.8239	0.9531	40.2764	592.5204
	PH: Equation (12)	2	0.3121	0.1867	0.5903	0.5284	7.7728
		3	0.3644	0.0418	0.4528	0.0768	1.1295
		4	0.3799	0.0092	0.3636	0.0144	0.2124
		5	0.3829	0.0019	0.3025	0.0027	0.0404

This table compares key stationary performance metrics for the M/M/c and fitted MAP/PH/c models, including idle probability, queue-formation probability, utilisation, and queuing delays across different numbers of connectors. The results illustrate how increasing station capacity reduces congestion and improves service efficiency.

Table 7. Theoretical and empirical computational complexity of the MAP/PH/c matrix-analytic evaluation for $m_a = 3$ and $m_s = 4$.

c	Block dimension $d(3, 4, c)$	Normalised $d(3, 4, c)^3$	Empirical runtime (s)
1	15	1	0.0331
2	45	27	0.0071
3	105	343	0.0320
4	210	2744	0.2983
5	378	16003	10.3477

To assess the computational cost of the fitting process, we implemented a standard EM-based procedure to estimate MAP and PH parameters on the same dataset, using the same hardware configuration. The EM algorithm for the 3-phase MAP and 4-phase PH models converged within 30-60 minutes with only 10 iterations, reflecting the well-known $O(TNm_s^2)$ scaling of an EM with a training sample size, N, number of service phases, m_s, and number of iterations, T (Parra and Barrett 1998). In contrast, the BuTools MAP and PH fitting routines completed in approximately 10.76 and 0.18 seconds, respectively, because they rely primarily on moment-based matrix operations. Given this substantial difference in runtime, EM-based fitting was used only as a benchmark, with all reported MAP and PH estimates being based on the more efficient BuTools calibration.

As shown in Table 7, the MAP/PH/c performance evaluation becomes highly efficient once the MAP and PH distributions have been estimated: computation completes in well under one second for up to four servers, and only increases to roughly 10 seconds at $c \geq 5$ runtime, which is consistent with the expected cubic growth in $d(m_a, m_s, c)$ described in Section 3.4. These results demonstrate that the proposed matrix-analytic approach remains fully tractable for realistic fast-charging configurations and can be scaled further when needed.

In general, the size of the phase is the primary source of computational burden in an MAP/PH/c model, as it grows alongside the number of arrival and service phases. Although matrix truncation can be applied, we found it had little effect on runtime because the dominant cost is governed by the block dimension rather than the number of QBD levels. In practice, tractability is maintained more effectively through dimensionality-reduction strategies, such as merging similar PH phases or selecting the number of MAP/PH phases through data-driven clustering, which prevents the unnecessary expansion of the state space. The use of BuTools further reduces fitting time and offers an additional practical efficiency gain when calibrating arrival and service models.

The numerical analysis presented in this section highlights the strong capabilities of our developed MAP/PH/c queuing model's strong ability to accurately capture the stochastic behaviour of EV charging stations, particularly Tesla Superchargers, under varying configurations. The results demonstrate how increasing the number of connectors and optimising the charging rate significantly enhance station performance by reducing congestion, minimising waiting times, and improving service reliability. The inclusion of a comparative M/M/c baseline further confirms that the proposed MAP/PH/c model provides a more realistic and conservative representation of system congestion under empirical charging behaviour. Additionally, the computational complexity results show that the model remains tractable for practical, fast-charging configurations, supported by efficient MAP/PH fitting and phase-selection strategies. The findings presented in this section consistently support the model's effectiveness in providing valuable insights into the dynamics of this system.

5. Managerial implications

The MAP/PH/c queuing model proposed in this paper offers a practical, realistic approach to efficient EV charging station management, as it addresses the limitations of overly simplistic traditional models. As EV adoption continues to increase, long queues and congestion at charging stations have emerged as pressing issues, necessitating a shift from simplistic models to more realistic frameworks that can accurately capture the complexities of real-world operations. Our model provides valuable insights into the optimisation of key factors affecting EV charging stations, such as station capacity, pricing strategies, and the appropriate number and type of connectors required to avoid congestion and improve service reliability. By incorporating this advanced modelling approach, Tesla Supercharger managers and other EV charging service providers can better align station operations with actual demand patterns observed in different locations, thus ensuring a seamless charging experience for users. This, in turn, fosters greater confidence among potential EV buyers, reducing range anxiety and encouraging wider EV adoption. Furthermore, the proposed model can be extended beyond stationary charging services; for example, it can be applied to optimise battery swap stations and to effectively distribute mobile charging units based on regional demand fluctuations. Ultimately, the adoption of this realistic queuing model can empower decision-makers to strategically plan and manage EV infrastructure, thereby improving user satisfaction and contributing to the broader transition toward sustainable transportation.

The dataset used in this study reflects actual EV users in the selected region; as such, certain models account for the bulk of the charging activity due to market adoption patterns. Tesla vehicles represent a large share of the EV fleet in many North American areas, while an expanding mix of BEVs and PHEVs continues to contribute to public charging records as adoption diversifies. Our dataset aligns with this reality, as it captures charging behaviour for vehicles representing multiple manufacturers and technologies. We relied on Tesla Supercharger data because Tesla provides publicly documented charging-rate characteristics across different state-of-charge levels, allowing us to interpret the PH phases in our model in a realistic and behaviour-consistent way. Although incorporating additional regions with broader vehicle diversity would further enrich the analysis, the MAP/PH/*c* framework developed in this study is fully generalisable and can be applied to any charging network once the relevant charging-rate characteristics are available. The methodological approach provided in this paper remains applicable, regardless of vehicle type or charging provider.

Although the proposed MAP/PH/*c* framework captures key stochastic features of EV charging behaviour, several real-world factors fall outside the scope of the current formulation. In particular, the model assumes that all arriving vehicles join the queue and remain until service completion, whereas actual charging stations may experience user impatience, such as customers deciding not to wait (balking) or leaving the queue before charging begins (reneging). Similarly, the model does not explicitly account for spatial capacity, which can restrict how many vehicles can queue at a station during peak hours. These behaviours are well recognised in queueing theory and can be integrated into the MAP/PH/*c* structure by introducing finite buffers, abandonment phases, or retrial mechanisms. Incorporating these extensions represents a natural direction for future research and would further enhance the model's realism and operational relevance.

This structured approach provides a quantitative, SoC-aware foundation for determining the PH distribution, and it offers a clear pathway for future extensions, such as estimating separate PH models conditioned on SoC intervals or allowing service rates to vary dynamically with SoC during charging.

While the proposed model provides a solid foundation for understanding and optimising EV charging infrastructure, future work could further enhance it by incorporating additional customer behavioural dynamics, such as customer impatience, retrials, and priority-based access through advanced booking systems. Furthermore, extending the model to account for stations with heterogeneous connectors would enhance its ability to address the diverse charging requirements of various EV battery types. Another promising direction for future research is the optimisation of pricing strategies by introducing penalties for vehicles remaining connected after reaching full charge. Such measures would help to mitigate congestion at high-demand locations (e.g. highway stations) during peak travel periods. These enhancements would provide a more comprehensive framework for managing EV charging stations, thus ensuring improved service efficiency and a seamless user experience as EV adoption continues to grow.

6. Conclusion

In this study, we developed a comprehensive MAP/PH/*c* queuing system to model EV charging stations, with a particular focus on Tesla Superchargers. This work represents a significant contribution to the field, as it is one of the first to implement such an advanced stochastic framework in the context of real-world EV charging operations. In addition, our thorough documentation of the theoretical and mathematical foundations of the embedded stochastic processes provides a robust framework that moves beyond simple queuing models. Furthermore, the proposed MAP/PH/c model was carefully evaluated based on key performance metrics and validated using numerical examples, thus ensuring its feasibility and accuracy in capturing system dynamics. This validation confirms the model's ability to account for the inherent variability of EV charging operations, allowing it to estimate congestion and station utilisation more precisely compared to simplistic Poisson-based models. In addressing the critical limitations of traditional queuing frameworks, the MAP/PH/c model asserts itself as a powerful tool capable of representing the intricate realities of EV charging systems.

Furthermore, by fitting MAP and PH distributions to empirical EV arrival and charging duration data, our model effectively captures the stochastic nature of real station operations and incorporates real-world complexities that are often simplified in theoretical studies. The findings confirm the model's ability to

provide actionable insights into station performance, congestion management, and infrastructure planning. This study marks a critical step toward bridging the gap between advanced theoretical queuing models and their practical implementation, offering a realistic foundation for optimising EV charging stations and addressing the growing demands of sustainable transportation systems.

City planners can leverage our model to design EV charging networks that minimise congestion and enhance service reliability, while charging station operators can utilise the model to optimise resource allocation, such as determining the optimal number and type of connectors required to meet varying demand across different locations. The proposed model and method can also help policymakers evaluate the effectiveness of policy measures (e.g. subsidies for EV infrastructure) and their impact on system performance and user experience. We also recognise that interpreting insights with user behaviours and multi-source data can involve cyber-security and data privacy methods, which opens future research directions. Finally, this work illuminates a number of new research avenues, such as integrating renewable energy sources, implementing dynamic pricing strategies, and addressing user behaviour complexities, including priority, impatience, and retrials.

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Author contributions

None.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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